DRISHTISCAN

Packaged-Commodity Compliance Scanner

Project Report and Technical Documentation

Purpose
Document the project, its workflow, implementation, verification logic, reporting system, deployment model, current limitations, and planned accuracy improvements.

Public source repository: github.com/bitsubhayu/dristiScan

Prepared as a supporting technical report for the project presentation.

1. Project Overview

DrishtiScan is a packaged-commodity compliance checking system designed to convert information printed on a product label into structured fields, evaluate those fields against applicable packaged-commodity requirements, and produce a human-readable compliance result. The system is intended to reduce the time and inconsistency involved in manually reading dense labels and checking long rule sets.

The central design goal is not to replace an officer or make an unquestionable legal decision. It is to turn a repetitive inspection workflow into a repeatable software workflow in which extracted evidence, rule outcomes, and uncertainty are visible and reviewable.

2. Problem Being Solved

- Packaged-commodity labels can contain many small, crowded, differently formatted declarations.
- Manual reading and transcription can introduce field-level errors.
- Checking every declaration against a long rule set is slow and can vary between inspectors.
- A wrong extraction is especially dangerous when it is displayed as a definite PASS rather than being surfaced for REVIEW.
- High inspection volume creates a need for faster, standardized first-pass checking.

3. Core Solution

DrishtiScan follows a pipeline in which a user supplies photographs of a product label. The system extracts text, identifies relevant declarations, normalizes them into structured fields, performs rule evaluation, performs consistency checks where applicable, and generates a compliance report.

#	Stage	What happens
1	Capture	User uploads one or more clear label photographs.
2	OCR extraction	PaddleOCR reads text from the supplied images.
3	Field mapping	Extracted text is classified into structured declarations such as product name, brand, quantity, prices, manufacturer, batch details, dates, origin, and consumer-care information.
4	Verification / reconciliation	Evidence from multiple photographs can be compared; uncertain or conflicting values can be re-checked using the project verification layer.
5	Rule engine	Structured fields are evaluated against packaged-commodity compliance rules implemented in the application.
6	Consistency checks	Cross-field relationships can be used to flag suspicious combinations instead of silently accepting them.
7	Report generation	Final values, findings, and evidence-oriented explanations are assembled into downloadable reports.

4. System Architecture

The project is organized as a web application with a frontend, backend, OCR processing, database persistence, rule evaluation, and report generation. OCR is kept as an independent service so that OCR workload can be scaled separately from the main application.

Layer	Technology / component	Role
Frontend	React + Vite	Web interface for upload, scan flow, results, and report access.
Backend	Node.js	API/orchestration layer connecting extraction, verification, rules, database, and reporting.
OCR	PaddleOCR	Primary text-recognition engine for label photographs.
Database	MongoDB	Stores application data and structured scan information.
Rule engine	Node.js rule logic	Maps structured declarations to compliance rules and findings.
Mismatch checking	Comparison module	Compares extracted values with externally supplied reference information when that workflow is used.
Reporting	PDF / DOCX generation	Produces formal, readable inspection outputs.

5. Data Extraction and Field Mapping

The extraction layer does more than return raw OCR text. It attempts to turn label text and spatial relationships into named fields. Examples include:

- Brand name
- Product / commodity name
- Generic commodity name
- Net quantity
- Serving size / servings information when present
- MRP
- Unit sale price
- Manufacturer name and address
- Marketer name
- Batch / lot number
- Dates such as manufacture and expiry
- Country of origin
- Consumer-care contact information

A key implementation concern is preserving enough spatial evidence to distinguish nearby but unrelated numbers. Flattened OCR text can create false associations, especially for fields such as servings and prices.

6. Compliance Rule Engine

After extraction, the system converts fields into rule inputs. The rule engine produces structured findings rather than returning only a single score. A finding can represent a PASS, FAIL, or REVIEW state depending on whether a declaration is present, compliant, uncertain, or internally inconsistent.

- Rules are implemented as application logic and linked to the relevant packaged-commodity requirements used by the project.
- The important design principle is traceability: a finding should be understandable from the extracted field and the rule that evaluated it.
- A REVIEW outcome is intentionally different from a failure. REVIEW is the safe state when the system cannot establish sufficient evidence.

7. Evidence, Verification, and Confidence Handling

The current implementation contains a verification layer intended to improve extraction robustness. The most important lesson from testing is that the integration policy matters as much as the verification model itself.

- A verification result must be allowed to replace a primary value when the primary value is empty or explicitly marked low-confidence.
- Verification should not blindly overwrite a high-confidence primary extraction.
- High-stakes fields should receive verification even when multiple photographs agree, because a consistent mistake can survive disagreement-based logic.
- Fields that originate from uncertain evidence should carry that uncertainty into the final report instead of being displayed identically to confident values.
- For visual verification, source-grounding checks should be used where possible; a value with no traceable supporting phrase should be treated cautiously rather than accepted automatically.

8. Accuracy Findings Identified During Testing

The following findings were traced to the project code and are important for the current accuracy overhaul.

Finding	What was observed	Required direction
Overwrite policy	Verification results were previously applied only when the existing field was empty. That prevents a correct verification result from replacing a wrong-but-present low-confidence extraction.	Use a confidence-based overwrite gate.
Disagreement-only verification	One reconciliation path was triggered only when candidates from different photos disagreed. A systematic mistake repeated in every photo could therefore	Always verify a selected set of high-stakes fields on each scan.

Finding	What was observed	Required direction
	escape that path.	
Field eligibility	Brand name and product name were absent from one low-confidence fallback field list, so existing verification descriptions were not actually reachable through that path.	Wire the fields into the fallback eligibility list.
MRP recovery	MRP was not wired into the same fallback path even though several OCR-text recovery strategies already existed.	Add MRP to the verification/fallback pathway and distinguish MRP from unit sale price.
Uncertainty propagation	A low-confidence unit-sale-price value could appear in a report without a visible uncertainty marker when no fallback value was applied.	Carry the source confidence state into the final report.
Cross-field validation	The code did not yet validate relationships between fields such as quantity, serving size, servings, prices, and dates.	Add post-merge consistency validators and downgrade suspicious fields to REVIEW.
Country-of-origin consistency	Different UI/report paths must use the same authoritative merged field so a value cannot appear differently in rule findings and declarations.	Introduce or enforce a single source of truth for displayed and evaluated values.

9. Cross-Field Consistency Validation

OCR errors are not always obvious from a single field. The stronger approach is to test whether related fields agree with one another after extraction.

- Net quantity / serving size should approximately agree with the reported number of servings, allowing for an explicit tolerance.
- MRP and unit sale price should be evaluated for implausible relationships, with appropriate unit normalization.
- Manufacture and expiry dates should be ordered correctly and should imply a plausible shelf-life interval.
- A failed consistency check should result in REVIEW rather than leaving the suspect value presented as unquestionably correct.

10. Reporting and Auditability

The reporting layer is part of the compliance workflow, not an afterthought. The output is intended to make the result usable by a person who needs to inspect, file, or explain the finding.

- Structured declaration values are shown in a readable form.

- Rule findings are linked to the corresponding extracted information.
- Uncertain or verification-assisted values are intended to be distinguishable from confident values.
- The system supports downloadable PDF and DOCX reports.
- The report should preserve enough context for a reviewer to understand why a field was marked PASS, FAIL, or REVIEW.

11. Multi-Photo Strategy

The scanner can use multiple photographs of the same commodity to improve robustness. The purpose is not simply to collect more OCR text; it is to compare candidate evidence and detect conflicts.

- Agreement across photos can increase confidence, but agreement alone does not prove correctness.
- Disagreement is a useful signal for reconciliation or human review.
- A repeated mistake across all photos must also be guarded against, which is why selected high-stakes fields require verification even without disagreement.

12. Deployment Model

The application is designed to run as separate web/API/OCR components so each workload can be deployed and scaled according to its resource requirements.

Component	Recommended role	Notes
Frontend	Static/web hosting	Public HTTPS URL used by users.
Node.js backend	Application hosting	Handles API requests, orchestration, rules, reporting, and database interaction.
OCR service	Independent service	PaddleOCR workload can be scaled separately from the API.
MongoDB	Managed database	Persistent application and scan data.
Public source	GitHub	Repository for project source and technical reference.

13. Public Repository

Project source repository: https://github.com/bitsubhayu/dristiScan

This repository is the public code reference associated with the project report.

14. Current Strengths

- End-to-end workflow from label photographs to structured compliance output.
- PaddleOCR is separated as an independent OCR workload rather than coupling OCR execution directly to the web interface.
- Field-specific extraction logic exists for important packaged-commodity declarations.

- A deterministic rule-engine layer provides explicit compliance findings instead of relying on an opaque overall score.
- Multi-photo processing creates opportunities for conflict detection and evidence comparison.
- PDF/DOCX report generation makes the result usable beyond the browser session.

15. Current Limitations and Risks

- Curved, damaged, low-contrast, reflective, or poorly photographed labels can reduce OCR reliability.
- An OCR engine can produce a plausible-looking but incorrect value; confidence handling and cross-field checks are therefore essential.
- Flattened OCR text can lose spatial relationships between nearby fields.
- Adding verification calls improves robustness but introduces latency and infrastructure/API considerations.
- A compliance system should prefer REVIEW over silently presenting an uncertain result as definite.
- Ground-truth testing must use the exact physical product/pack variant being evaluated; otherwise a comparison against a different pack size can produce a false bug report.

16. Accuracy Target and Evaluation Philosophy

For this type of compliance tool, overall raw OCR accuracy is not the only useful metric. The most important safety metric is false confidence: how often the system presents an incorrect field as a definite result rather than marking it REVIEW.

- Optimize for a very low false-confidence rate.
- Allow a meaningful REVIEW rate when the evidence is weak or conflicting.
- Measure field-level correctness and confidence separately.
- Test the same physical products repeatedly after changes so regressions are visible.

17. Verification Plan

1. Re-run the same controlled product photographs after every extraction/verification change.
2. For each important field, record primary OCR extraction, verification output, final merged value, rule finding, and final rendered report value.
3. Use a confirmed ground-truth fixture for pack size and servings before declaring a defect.
4. Check that low-confidence values are visibly treated as uncertain in the final output.
5. Check that arithmetic/date validation can downgrade a field to REVIEW.
6. Check that the value shown in declarations and the value used by the rule engine come from the same authoritative field.

18. Recommended Next Engineering Priorities

1. Implement the confidence-based overwrite gate for verification results.

2. Run verification on brand, product name, generic commodity name, MRP, and unit sale price on every scan.

3. Complete the verification/fallback wiring for brand, product name, and MRP.

4. Add cross-field arithmetic/date validators after the merge stage.

5. Enforce a single source of truth for fields used in both reports and rule evaluation.

6. Add a stage-by-stage diagnostic view for test mode so extraction errors can be traced to the exact pipeline stage.

19. Project Positioning

DrishtiScan should be presented as a practical compliance scanner and decision-support workflow for packaged commodities. Its value comes from combining OCR, structured field extraction, deterministic rule evaluation, evidence comparison, confidence-aware handling, and formal reporting into one repeatable process.

The system is strongest when it makes uncertainty explicit. A result that says REVIEW because the evidence is weak is more defensible than a fast but incorrect PASS.

Source note: This report is based on the project information available in the conversation and the attached code-traced accuracy analysis. The supplied presentation was intentionally not used as the source for project claims in this report.